CNS

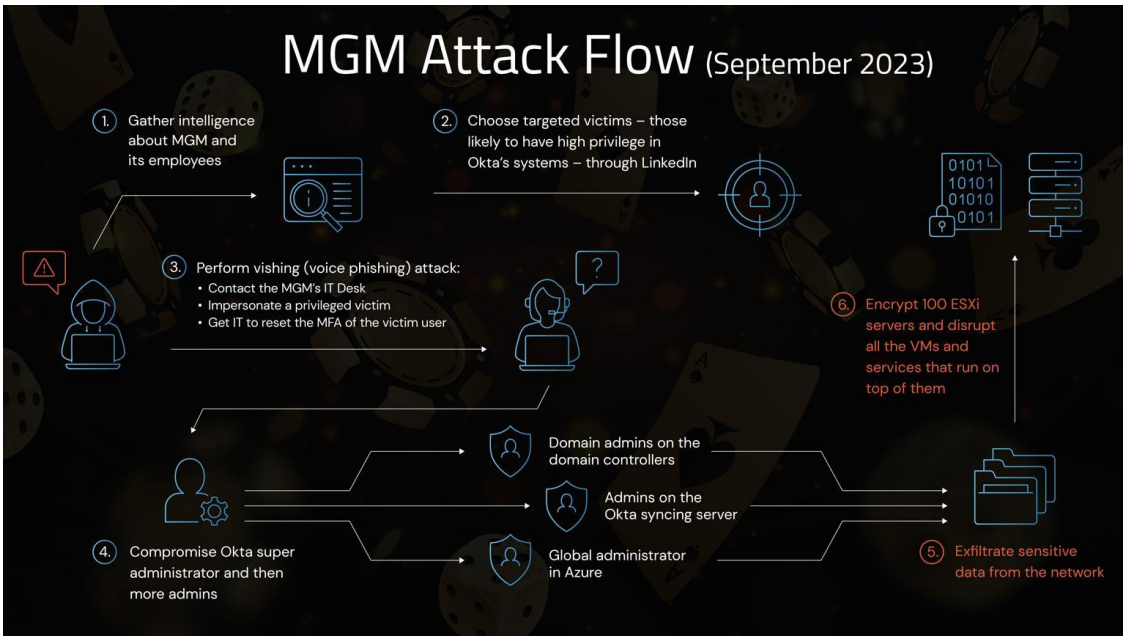
ALA-3

AIM: In this activity, students will create an infographic on Social Engineering Attacks and Major Security Breaches, highlighting real-world case studies, attack techniques (phishing, pretexting, baiting), and prevention strategies. The project should include engaging visuals, flowcharts, and brief analyses of how these attacks occurred and how they could have been prevented. Students will submit the presentation file and screenshot of the final design on the GMIU Web Portal.

Solution:

➢ **5 Real-World Cybersecurity Case Studies**

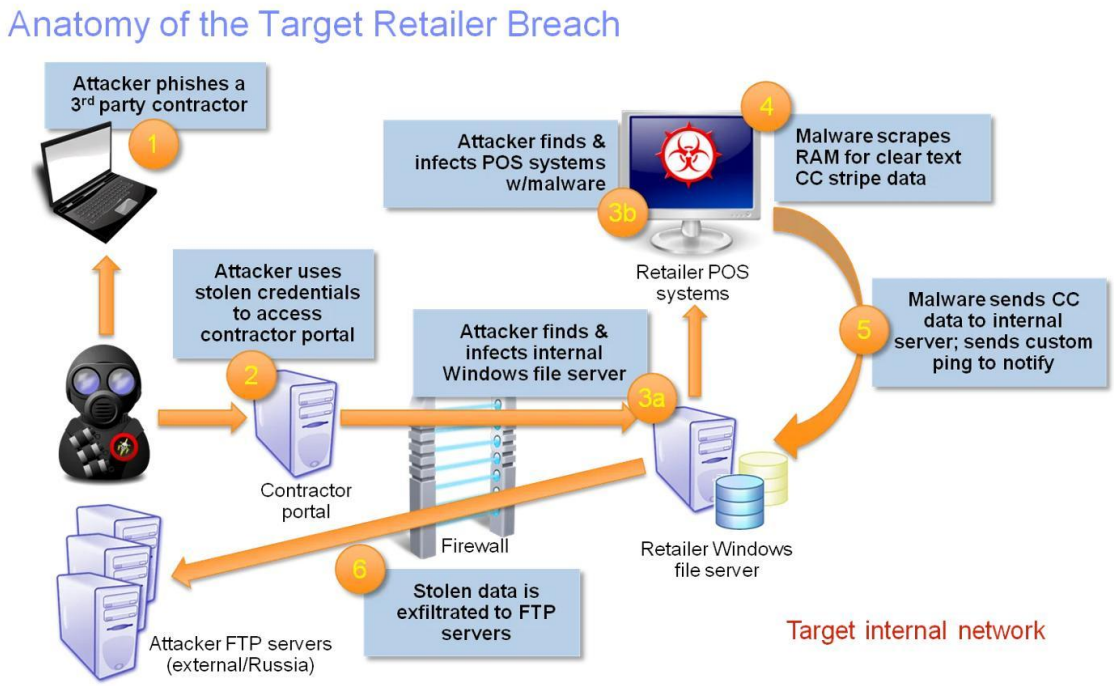
1. MGM Resorts Cyberattack (2023)



- Attack type: Social engineering and help-desk impersonation.
- What happened: Attackers reportedly used social engineering to gain access to MGM Resorts' systems, disrupting hotel reservations, casino operations, and digital services.

- Impact: Major operational disruption and financial losses.
- Prevention: Verify employee identities, strengthen help-desk procedures, and use multi-factor authentication (MFA).

2. Target Data Breach (2013)



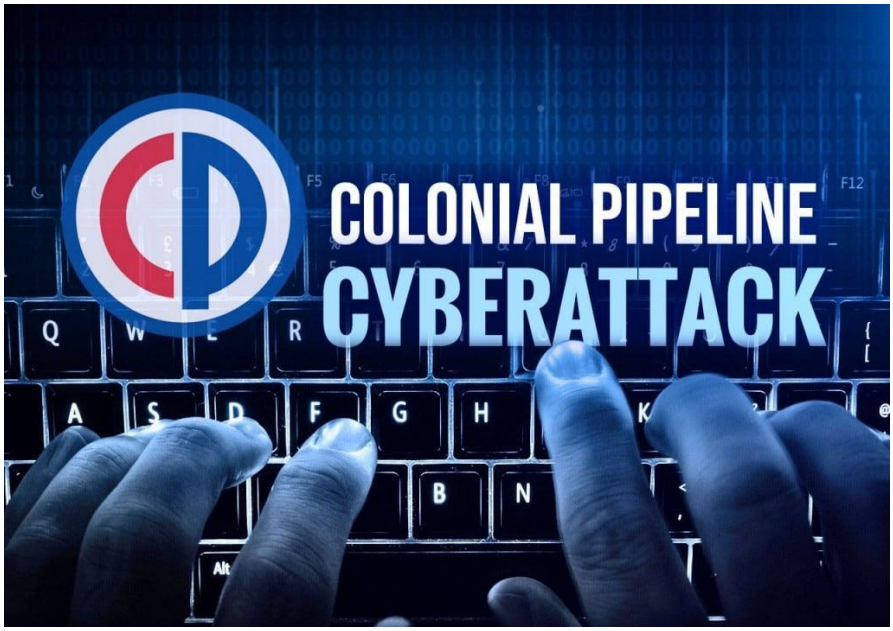
- Attack type: Stolen third-party vendor credentials.
- What happened: Attackers obtained credentials associated with a Target vendor and used them to gain access to Target's network.
- Impact: Payment-card information belonging to millions of customers was compromised.
- Prevention: Restrict vendor access, monitor unusual activity, and use network segmentation.

3. Twitter (X) Cyberattack (2020)



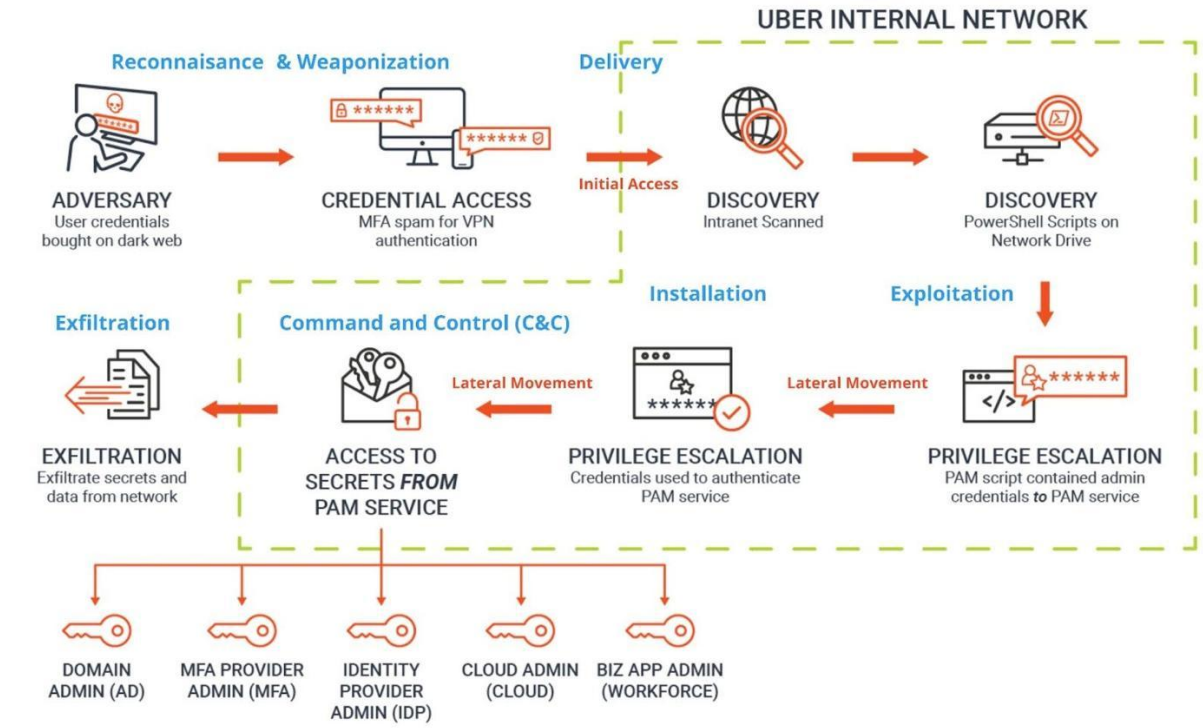
- Attack type: Phone phishing and impersonation.
- What happened: Attackers pretended to be IT support and manipulated employees into providing access to internal systems.
- Impact: They took control of high-profile accounts and posted fraudulent Bitcoin messages.
- Prevention: Verify IT support requests, restrict internal access, and train employees to recognize social engineering.

4. Colonial Pipeline Ransomware Attack (2021)



- Attack type: Ransomware using a compromised password.
- What happened: Attackers accessed Colonial Pipeline's network using a compromised password for a remote-access account.
- Impact: The company temporarily shut down pipeline operations, disrupting fuel supplies in parts of the United States.
- Prevention: Use multi-factor authentication, disable unused accounts, and monitor remote access.

5. Uber Cyberattack (2022)



- Attack type: Social engineering and MFA fatigue.
- What happened: An attacker reportedly obtained a contractor's login credentials and repeatedly sent authentication requests, eventually persuading the contractor to approve access.
- Impact: The attacker accessed internal systems and several company tools.
- Prevention: Never approve unexpected login requests, use phishing-resistant MFA, and immediately report suspicious authentication activity.